

ASHWIN T E

Mechatronics & Automation | Robotics | Embedded Systems

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PROFILE

Mechatronics & Automation undergraduate focused on autonomous systems, embedded firmware, and robotics. Hands-on experience with ROS2, SLAM, ESP32/STM32, RTOS, custom PCB/PDB design, and computer vision. Led and advised competition teams at NASA HERC while developing patent-pending agricultural and medical-device systems.

EDUCATION

Vellore Institute of Technology (VIT), Chennai

2024 - Present

B.Tech - Mechatronics and Automation

TECHNICAL SKILLS

Robotics & Control: ROS2, SLAM, path planning, kinematics, control systems

Embedded & Firmware: Embedded C/C++, ESP32, STM32, Raspberry Pi 5, Jetson Nano, RTOS, Arduino

Vision & Programming: Python, OpenCV, TensorFlow/PyTorch, Java, Assembly, SQL

Hardware & Design: PCB/PDB architecture, SolidWorks, Fusion 360, LTspice, CAD/CAM, DFM, VMC

EXPERIENCE & LEADERSHIP

AutoVIT Club - Secretary

2024 - Present

- Lead technical operations, team coordination, workshops, and autonomous-vehicle research across student mechatronics projects.

Team MOVIS - Advisory Board Member | Former Programming Lead & HERC POC

2025 - Present

- Led programming and electrical implementation for NASA HERC 2026; now advise software/electrical architecture for autonomous rover systems.

MEL Systems and Services - Embedded System Design Intern

May - Jun 2025

- Worked on the Pick2Lite hardware/IoT product, contributing to embedded logic and electronic system architecture.

Salcomp Manufacturing - Engineering Intern

May - Jun 2026

- Applied CAD modeling and DFM principles and gained hands-on exposure to VMC manufacturing operations.

SELECTED PROJECTS & PATENTS

Agri-Sorter - Patent Pending

- Developing an automated fruit-quality sorting system combining ultrasonic sensing, computer vision, and mechatronic size-based classification.

Portable Air-Puff Tonometer - Patent Pending

- Developing a non-contact intraocular-pressure measurement device using controlled micro-pneumatics and embedded sensing.

ISRO Robotics Challenge - Autonomous Drone

- Built autonomous-flight logic and motor-control integration using ROS2, Raspberry Pi, and Pixhawk.

Autonomous Delivery Robot

- Developed SLAM-based navigation and obstacle detection; designed the electronics architecture including custom PDB/PCB hardware.

SELECTED ACHIEVEMENTS

- NASA Human Exploration Rover Challenge 2026 - 6th Place Global (RC division), Best Indian Team, 7:02 fastest run.
- REVIVE 2024 - Winner, National-Level Hackathon.
- Innovate National Hackathon - 3rd Place.
- Caterpillar Autonomy Challenge (IIT Madras & Caterpillar) - Semi-finalist in autonomous navigation.
- BITS Hyderabad Project Expo - Finalist.

Languages: English | German | Tamil | Hindi